

ITERATION 1 (Constantine)

Starting state $m = [-0.9314, 1.9657]$
 $b = [1.0001, 1.00085]$

Compute Predictions

$$\hat{y}_1 = m_{1,20} \times m_{1,2} \times 5 + b_1$$

$$= (-0.9314 \times 1) \times (1.9657 \times 5) + 1.0001$$

$$= -4.492458940000001$$

$$\hat{y}_2 = (-0.9314 \times 1.9657) \times (4 \times 10) + 1.00085$$

$$= -72.2332692$$

Compute Errors:

$$e_1 = -4.492458940000001 - 5 = -9.49245894$$

$$e_2 = -72.2332692 - 6 = -78.2332692$$

Compute Gradient

$$\frac{dJ}{dm_1} = m_2 (e_1 + 40e_2) = 1.9657 \times (3x - 9.49245894 + 40 \times -78.2332692)$$

$$= -6207.303470272674$$

$$\frac{dJ}{dm_2} = m_1 (e_1 + 40e_2)$$

$$= -0.9314 \times 3157.8081448199916$$

$$\frac{dJ}{dm_2} = 2941.1825060853475$$

$$\frac{dJ}{db_1} = e_1 = -9.49245894$$

$$\frac{dJ}{db_2} = e_2 = -78.2332692$$

Update Parameters

$$m_1 = -0.9314 - 0.0001 \times (-6207.303470272674)$$

$$m_1 = -0.8693269652972733$$

$$m_2 = 1.9657 - 0.0001 (2941.1825060853475)$$

$$= 1.9362881749391465$$

$$b_1 = 1.0001 - 0.0001 (-9.49245894) = 1.0001949245894$$

$$b_2 = 1.00085 - 0.0001 (-78.2332692) = 1.000682332692$$

Updated state after iteration 1

$$m_2 = [0.8693269652972733, 1.9562881747391465]$$

$$b_2 = [1.0001949245894, 1.001632332692]$$

compute sigma

$$d + \sum_{i=1}^n m_i \times \sum_{j=1}^n m_j = \hat{\mu}$$

$$1000 \cdot 1 + (2 \times 10^5 \cdot 1) \times (1 \times 10^5 \cdot 0) = \hat{\mu}$$

$$100000 + 10000000 \cdot 0 = \hat{\mu}$$

$$28000 \cdot 1 + (0.1 \times 10^5) \times (0.2 \times 10^5 \cdot 0) = \hat{\mu}$$

$$28000 + 2000000 \cdot 0 = \hat{\mu}$$

compute sigma

$$1000000 + 10000000 \cdot 0 = \hat{\mu}$$

$$28000 + 2000000 \cdot 0 = \hat{\mu}$$

compute sigma

$$(1000 + 1000000) \times (0.2 \times 10^5) = \hat{\mu}$$

$$1000000 + 200000000 = \hat{\mu}$$

$$28000 + 200000000 = \hat{\mu}$$

$$(1000 + 1000000) \times (0.2 \times 10^5) = \hat{\mu}$$

$$1000000 + 200000000 = \hat{\mu}$$

$$28000 + 200000000 = \hat{\mu}$$

$$1000000 + 100000000 = \hat{\mu}$$

$$28000 + 200000000 = \hat{\mu}$$